

Paper Manager pack — Data Scaling Laws (Nov 23 · Data)

Paper: Data Scaling Laws in Imitation Learning for Robotic Manipulation
arXiv: <https://arxiv.org/abs/2410.18647> · [project](#) · [code](#) · [HF data](#) · ICLR 2025 Oral
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PDF: `/workspace/papers/data-scaling-laws-2410.18647.pdf` (= `/workspace/data-scaling-laws-quiet/paper.pdf`) · 34 pp · arXiv **2410.18647v4**
Session: Nov 23 · due 9:30 AM · Type: **Data**
Role weights: Empiricist=3 (Nykolas Rekasius) **LONGEST PRIMARY**; other seats empty — **still all five produced**; Empiricist heaviest
Kanta role: Produce **Empiricist-richest** five-role pack; deepen diversity-scaling spine + motion/policy notes + mini-N reproduce checklist (RC0–RC5)
Themes: data scaling laws · env/object/pair diversity $\gg$ demo depth · power-law OOD · UMI $\rightarrow$ Franka · Diffusion Policy + ACT temporal ensemble · DINOv2 ViT capacity · $K \approx 50$ plateau · 32×50 recipe
Sources: Summarizer (`/workspace/papers/data-scaling-laws-2410.18647-summary.md`) + Senior Research (`/workspace/data-scaling-laws-quiet/DELIVERABLE.md`) + Ablation Bot (`/workspace/papers/data-scaling-laws-2410.18647-ablations.md` — **AUTHORITATIVE**) + `paper_notes.md` / `env_outline/`
Sibling links: `/workspace/umi-quiet/` · `/workspace/diffusion-policy-quiet/` · `/workspace/droid-quiet/` · `/workspace/oxe-quiet/` · `/workspace/act-aloha-quiet/` · `/workspace/openvla-quiet/` · `/workspace/pi05-quiet/` · `/workspace/openpi-comet-smoke/` · `/workspace/dart-quiet/` · `/workspace/robogen-quiet/`

1. One-liner

Diversity (envs / objects / pairs) $\gg$ demo depth — for single-task IL, OOD gap follows **power laws** in $\backslash (M, N, \text{pairs})$; demos/pair **plateau** ~ 50 ; recipe $\backslash (M \{=\} N \{=\} 32 \backslash)$ **pairs** $\times \backslash (K \{ \approx 50 \}) \rightarrow \sim 90\%$ zero-shot on novel env+object (DP + DINOv2 FT + ACT TE).
Contrast poles:

Pole	What they scale	Primary claim
This paper (Lin'25)	Envs $\times$ objects $\times$ demos (single-task DP)	Power-law OOD env/object gen; diversity $>$ demo depth
OXE	Embodiments $\times$ labs $\times$ skills	Positive transfer across robots; foundation mix
DROID	Scenes (same Franka)	Scene diversity ablation (+OOD); co-train recipe
UMI	Handheld interface + in-wild locations	Collection <i>mechanism</i> this paper uses
Octo / OpenVLA / π_0	Multi-task generalist + model/compute	Task-level + language; this paper excludes task scaling

Do NOT invent: universal multi-task / multi-embodiment Chinchilla laws — paper fits **data axes** $\backslash (M, N, K)$ only inside single-task IL + DP.

2. Summary (exec overview + key numbers)

Lin et al. (ICLR 2025 Oral) ask whether **data scaling alone** can yield **single-task** imitation policies that deploy **zero-shot** to any same-category object in any environment. They collect **>40 000** UMI handheld demos, train **Diffusion Policy + DINOv2 ViT-L/14** (full FT) + **ACT temporal ensemble**, and run **>15 000** real Franka rollouts under a **stage-scored** protocol.
Headline recipe (CLEAR §4.3 / §5): collect in **as many environments as possible**, **one unique object per env**, $\backslash (K \{ \approx 50 \})$ demos/pair; at **32 pairs** (~ 1600 demos/task) expect $\sim 90\%$ zero-shot on unseen env+object — validated

on Pour Water, Mouse Arrangement, Fold Towels, Unplug Charger (Fold+Unplug: **one afternoon / four collectors**).

Why Data session: Supplies the data cluster's clearest **quantitative** answer to "how much data?" — **scenes/objects, not more demos of the same table.**

Axis	Number / claim		
Study scale	>40k demos · >15k real rollouts · 4 tasks · ~90% demos valid post-SLAM		
Power-law form	$\backslash(Y=1-S=\backslashbeta X^{\backslashalpha})$; $\backslash(X\text{in}\{N,M,\text{pairs}\})$; no $\backslash(Y_{\infty})$ (only 6 pts)		
PW exponents (obj / env / pairs)	−0.703 / −0.844 / −0.579 ($\backslash$	r $\backslash$	0.960–0.987)
MA exponents (obj / env / pairs)	−0.697 / −0.466 / −0.683 ($\backslash$	r $\backslash$	0.942–0.991)
MA → score 0.99	1,191 pairs (extrapolation; unverified)		
Demo vs score $\backslash(r)$	PW −0.62; MA −0.79 — no clear demo power-law		
Plateaus	~800 total @ M=16,N=64; 400/800/1600 for 8/16/32 pairs → K=50		
Object gen @ 8 / 32 objs	score >0.8 / >0.9		
Env gen moral	Harder than object (shallower early slope)		
Multi-obj @ M=16	Negligible vs 1 obj/env (Fig. 6)		
Table 1 SR (32×50)	Pour 85% · Mouse 92.5% · Fold 87.5% · Unplug 90%		
Table 1 scores	0.922 / 0.933 / 0.95 / 0.887 ($\pm$ CIs)		
ViT-S / B / L	0.66 / 0.81 / 0.90		
U-Net S / B / L	0.88 / 0.90 / 0.83 (scale hurts)		
FT / LoRA / LfS / frozen	0.90 / 0.72 / 0.03 / 0.00		
Params / max train	>396M; 5e5 steps; ~75 h on 8×A800		
Policy stack	DP CNN U-Net + DDIM-16; Ah=16 @ 5 Hz; TE=8 @ −0.01; BF16		

Must-cite tattoo: Fig.5 exponents · **K=50** plateau · Table1 **85–92.5%** · ViT **0.66/0.81/0.90** vs U-Net **0.88/0.90/0.83** · frozen/LfS/LoRA **0.00/0.03/0.72** · MA → 0.99 needs **1191** pairs · diversity >> depth · **32×50**.

Do NOT claim: multi-task/language laws; multi-embodiment Chinchilla; course smoke = Tab.1 ~90%; 1191-pair verified; MSE replaces human scores.

3. Keywords

data scaling laws; imitation learning; robotic manipulation; power law; optimality gap; environment generalization; object generalization; env–object pairs; demonstration diversity; UMI; Diffusion Policy; DINOv2; ACT temporal ensemble; Franka; zero-shot deployment; Kaplan analogy; ICLR 2025 Oral; K=50; 32×50 recipe; BEHAVIOR data budget; ViT vs U-Net; LoRA vs FT

4. Deep dive — Diversity scaling (PRIMARY DATA CONTRIBUTION)

Callout: Scientific product = **empirical power laws on diversity axes** for single-task IL OOD — not a foundation model, not a planner, not a VLA. Session type = Data; quantitative backbone for "spend on scenes/objects, not depth."

Formal axes (§3) — CLEAR

Symbol	Meaning
$\backslash(M)$	# training environments
$\backslash(N)$	# training manipulation objects (same category)
$\backslash(K)$	# demos per env–object pair
$\backslash(S)$	Normalized test score on unseen envs and/or objects
$\backslash(Y)$	Optimality gap $\backslash(Y=1-S)$ (fit target)

Power-law form (CLEAR §4.2): $(Y=\beta \cdot X^{\alpha})$ with $(X \in \{N,M,\text{pairs}\})$. Log-linear OLS. Footnote: 3-param $(Y=\beta X^{\alpha}+Y_{\infty})$ **not** fit — only 6 points.

Fig. 5 power-law fits (must-cite) — VERIFIED

Task	Axis (X)	Fit	Pearson (r) (log–log)		
Pour Water	# objects	$(Y=0.825 \cdot X^{-0.703})$	−0.987	r	$=0.987$
Pour Water	# envs	$(Y=1.180 \cdot X^{-0.844})$	−0.960		
Pour Water	# pairs	$(Y=1.068 \cdot X^{-0.579})$	−0.980		
Mouse Arrangement	# objects	$(Y=0.826 \cdot X^{-0.697})$	−0.991		
Mouse Arrangement	# envs	$(Y=0.827 \cdot X^{-0.466})$	−0.966		
Mouse Arrangement	# pairs	$(Y=1.263 \cdot X^{-0.683})$	−0.942		

pdftotext trap: Fig.5 exponents are **negative** (gap!); Ablation Bot corrected. Mouse pairs $\rightarrow (Y=0.01)$ (score 0.99) $\Rightarrow (X \approx \mathbf{1191})$ (CLEAR claim; **unverified** experimentally).

Diversity sweeps (100% demo norms) — Ablation Bot AUTHORITATIVE

Sweep	Pour Water (M/N/pairs 1 → 32)	Mouse Arrangement
Objects Fig.2	0.133 → 0.939	0.217 → 0.929
Envs Fig.3	0.144 → 0.956	0.217 → 0.867
Pairs Fig.4	0.050 → 0.875	0.125 → 0.921

Morals: (1) Env gen **harder** than object. (2) With more diversity, fewer demos/unit needed. (3) Joint diversity saturates demo-fraction curves fastest. (4) App.G.2: diversity still wins at **matched total demos**.

Demo saturation (Fig. 7) — NEGATIVE power-law

Setting	Plateau (~)
Max pool $(M=16,N=64)$	~800 total demos
8 / 16 / 32 pairs	~400 / ~800 / ~1600
Recommended (K)	50 demos / pair (similar difficulty)

Demo–performance (r) : PW −0.62, MA −0.79 — **weak**; more hours on same (env,obj) $\neq$ more OOD.

Collection recipe (§4.3 → Table 1)

- Maximize #envs; 1 unique object per env.
- $(M=N=32)$ pairs $\times (K \approx 50) \rightarrow \sim 1600$ demos/task.
- Multi-obj/env helps only when (M) **small**; at $(M=16)$ gap ≈ 0 (Fig. 6).

Task	Norm. score	Success rate
Pour Water	(0.922 ± 0.075)	85.0±19.4%
Mouse Arrangement	(0.933 ± 0.088)	92.5±9.7%
Fold Towels	(0.95 ± 0.062)	87.5±17.1%
Unplug Charger	(0.887 ± 0.14)	90.0±14.1%

Variance warning: Table 12 — e.g. Pour Water env#2 only 40% SR while others 80–100%.

Capacity (Table 2) — Pour Water, 32 pairs, 50% demos

Ablation	Scores
Train strategy	FT 0.90 · LoRA 0.72 · LfS 0.03 · frozen 0.00
Visual encoder	ViT-S 0.66 · ViT-B 0.81 · ViT-L 0.90
Action U-Net	small 0.88 · base 0.90 · large 0.83

Takeaway: vision pretrain + **full FT** + encoder size matter; action U-Net scale **does not** (may hurt). **LoRA MSE trap:** LoRA can win MSE yet lose stage score — primary = human/stage proxy.

Diversity exam bite: "OOD gap $\backslash(Y=1-S\backslash\approx\backslashbeta X^{\backslash\alpha}\backslash)$ with $\backslash(\backslash\alpha\backslash\sim\backslash0.5\backslash)$ to $\backslash(-0.8\backslash)$ on $\#envs/\#objects/\#pairs$; demos plateau (~50/pair); recipe **32×50** → ~90% ZS; diversity >> depth; **not** a Chinchilla multi-task law."

5. Deep dive — Motion / policy / control notes

Callout: Motion is **supporting**, not classical planner / not VLA primary. Stack = **Diffusion Policy** (chunked visuomotor diffusion) + **ACT temporal ensemble** for smooth chunk stitching. Embodiment = **single** Franka deploy; collection = **human+UMI**.

Policy / control stack (§3, App. C/F) — CLEAR

Knob	Paper default
Policy family	Diffusion Policy CNN 1D U-Net noise predictor
Sampler	DDIM, 16 inference steps
Vision	DINOv2 ViT-L/14, fully fine-tuned
Action smoothing	ACT temporal ensemble: ens steps=8, adaptation -0.01
Obs resolution	224×224 wrist RGB (GoPro Hero 10 fisheye)
Action horizon	16 @ env freq 5 Hz → ~3.2 s chunk
Obs horizon	2 default; 3 for Pour (0.25 s) / Unplug (0.5 s) distant history
Optim	AdamW; lr action 3e-4, encoder 3e-5; batch 256; BF16
Deploy	Franka Panda + Weiss WSG-50; soft 95A TPU fingers; NVIDIA 4090

Motion / LH hooks (Empiricist / BEHAVIOR)

Hook	Where	Use
Handheld UMI teaching	Sec.3	Diversity collect without robot fleet — /workspace/umi-quiet/
Relative EE + DP chunks	Policy	Ah=16 @ 5 Hz; TE=8 jerky-switch fix
Distant obs history	App.C	LH / stage-ambiguity ≠ more demos
Multi-stage scoring	App.D	Stage-wise SR for LH credit (0–3 pts/step)
Dynamic unplug	Unplug Charger	Speed / yank; UMI latency sibling
Non-prehensile push	Mouse	Push-before-grasp stage
Precision pour	Pour Water	Fine motion + history
Fold deformation	Fold Towels	Soft-object LH
Sibling stacks	DP / ACT / $\pi_{0.5}$ / OpenVLA	Keep diversity mix cards ; swap head

Implication: Env/object **pair-count**, **demos/pair**, and **vision capacity** are **separate knobs** (P1/P2/P5/P8) — **do not** collapse into "more demos."

Motion exam bite: "Lin'25 = **DP + DINOv2 FT + ACT TE** on UMI → Franka; not classical planner, not VLA primary; Ah=16@5Hz, TE=8; distant history for Pour/Unplug; diversity axes are the science."

6. Role-ready sections (all five — Empiricist RICHEST)

6.1 Empiricist — WEIGHT 3 LONGEST PRIMARY (Nykolas Rekasius)

Seat charge: Course smokes that stress $\#envs \cdot \#objects \cdot demo\ K \cdot power\text{-}law\ fit \cdot capacity$, ranked by GPU honesty. Prefer ruthlessly mini-N $M, N \in \{1, 2, 4\}$, $K \in \{10, 25, 50\}$ + toy power-law + tiny ViT-S vs B — not paper $M=32 / 8 \times A800 / 15k$ rollouts / Tab.1 ~90%.

Claim under measurement

Single-task Diffusion Policy OOD success on new envs/objects is a **power-law function of diversity counts**, not of raw demonstration count past a modest $\backslash(K)$.

Hypotheses (falsifiable — from DELIVERABLE)

- H-env (P1):** Held-out env stage-proxy rises **monotonically** with $M \in \{1, 2, 4\}$ at fixed object + nested demos (Pour $0.144 \rightarrow \dots \rightarrow 0.956$ moral; env harder than object).
- H-obj (P2):** Held-out object proxy rises with $N \in \{1, 2, 4\}$ at fixed env (Pour $0.133 \rightarrow \dots \rightarrow 0.939$).
- H-demo-sat (P5):** At fixed small $\#pairs$, $K \in \{10, 25, 50\}$ shows **diminishing returns**; no strong power-law vs K alone.
- H-power (P4):** Gap $\backslash(Y=1-\mathrm{score})$ vs $\backslash(X \ln \backslash\{M, N, \mathrm{pairs}\})$ approx log-linear with **negative** $\backslash(\alpha)$; demo-total alone is weak.
- H-vit (P8):** Under matched mini-N + steps, **ViT-B > ViT-S** on OOD proxy (0.81 vs 0.66 moral).
- H-unet-flat (P10):** U-Net base $\succ$ small on OOD; large may hurt (defer large).
- H-ft (P9):** Full FT $\gg$ LoRA on **stage proxy** even if MSE ranks LoRA better.
- H-div>vol (P7):** At matched total demos, higher M or N beats low-diversity high-K.
- H-recipe (P11):** Mini analogue of high-pair / $K \approx 50$ beats same total demos on ≤ 2 pairs for joint OOD — **without** claiming Tab.1 90%.

Falsifiers: $M=4 \succ M=1$ on env-OOD; N flat; K linear power-law with high r; ViT-S $\approx$ ViT-B; U-Net large $\gg$ base; LoRA $\geq$ FT on stage proxy.

Prioritized ablation matrix (Ablation Bot AUTHORITATIVE P1–P12)

Pri	Experiment	Paper hook	Smoke	Est. GPU-h
P1	$\#envs\ M \in \{1..32\}$	Fig.3 / Tab.5,9	Y — RC1 PRIMARY	4–14
P2	$\#objects\ N \in \{1..32\}$	Fig.2 / Tab.4,8	Y — RC2 PRIMARY	4–14
P3	$\#env\text{--}object\ pairs$	Fig.4 / Tab.6,10	Partial after P1+P2	4–12
P4	Power-law fit gap vs M/N/pairs	Fig.5	Y — RC4	<1–2 CPU
P5	Demo quantity K	Fig.7 NEGATIVE	Y — RC3 PRIMARY	4–12
P6	Objects-per-env heatmap	Fig.6	Partial	4–10
P7	Constant-total-demo replot	App.G.2	Partial	2–8
P8	Encoder ViT-S/B/L	Tab.2b	Y — RC5 (S vs B)	6–16
P9	Encoder strategy FT/LoRA/LfS/frozen	Tab.2a	Fold into RC5	—
P10	U-Net size	Tab.2c	Y — RC5	(shared)
P11	Cross-task recipe 32×50	Tab.1	Partial — RC6	8–20
P12	MSE power-law	App.G.1	N / light	<1
Defer	Full $M=32 / 1191\text{-}pair / 8 \times A800 / 15k$	—	N	$\gg 200$

Min package (Ablation Bot): P1 + P2 + P5 $\rightarrow$ P4 $\rightarrow$ P8/P10 $\rightarrow$ P11.

Default course smoke order:

shard_ingest → env_diversity_sweep (P1) → object_diversity_sweep (P2) → ndemo_sweep (P5) → powerlaw_fit (P4) → capacity_x_data (P8/P9/P10) → optional P3/P7/P11.

Exact reproduce stack (EDIT)

```
# Official
git clone https://github.com/Fanqi-Lin/Data-Scaling-Laws
# Project: https://data-scaling-laws.github.io/
# HF: https://huggingface.co/datasets/Fanqi-Lin/Processed-Task-Dataset
# Pin commits → logs/pins.txt

# Course scout env (quiet outlines – preferred)
export DSLAW_WORK_ROOT=$HOME/dslaw-smoke
cd /workspace/data-scaling-laws-quiet
bash env_outline/setup_mamba.sh
# mamba activate dslaw-scout
mkdir -p logs runs mix_cards configs
# EDIT partition/account in slurm_*.sh
# Outlines only – do NOT sbatch from quiet box
# NEVER claim Tab.1 ~90% / 1191-pair / paper-scale from mini-N
```

Run cards (RC0–RC6) — Empiricist LONGEST

RC0 — shard_ingest_smoke (CPU-first gate)

Field	Spec
Goal	Demo → tensors + diversity card schema (env_id, obj_id, pair_id)
Input	Public release shard or synthetic: ≥ 4 env $\times$ ≥ 4 obj $\times$ ≤ 50 demos; wrist RGB stub; relative EE+grip; stage labels
Pass	Schema OK; 0 NaNs; mix card lists M,N,K; wall <30 min CPU
Fail	Missing env/obj tags; abs labeled as rel; silent pair drops
Script	env_outline/slurm_shard_ingest_smoke.sh

RC1 — env_diversity_sweep (P1 PRIMARY)

Field	Spec
Paper	Pour norm $\approx 0.144 \rightarrow 0.956$; Mouse $\approx 0.217 \rightarrow 0.867$; env harder
Goal	Monotonic $\uparrow$ env-OOD with mini ME{1, 2, 4} (directional)
Train	Small DP (ResNet18 or ViT-S); 1 seed; nested envs (App. C)
OOD	Hold out ≥ 2 envs never in train; same object category
Pass	3 M runs; $M=4 \geq M=2 \geq M=1$ on env-OOD; plot — not 0.956
Budget	~4–14 GPU-h; VRAM 10–20 GB
Script	env_outline/slurm_env_diversity_sweep.sh (#SBATCH -a 0-2)
Cross-link	/workspace/droid-quiet/ scene-diversity; /workspace/umi-quiet/ wild vs narrow

RC2 — object_diversity_sweep (P2 PRIMARY)

Field	Spec
Paper	Pour $\approx 0.133 \rightarrow 0.939$; Mouse $\approx 0.217 \rightarrow 0.929$; 8 objs >0.8, 32 >0.9
Goal	Monotonic $\uparrow$ object-OOD with mini NE{1, 2, 4}
Pass	$N=4 \geq N=2 \geq N=1$ on object-OOD; steeper early than RC1
Budget	~4–14 GPU-h
Script	env_outline/slurm_object_diversity_sweep.sh

RC3 — `ndemo_sweep` (P5 PRIMARY)

Field	Spec
Paper	NEGATIVE power-law vs demos; plateau ~800; recommend K=50
Goal	Diminishing returns on $K \in \{10, 25, 50\}$ at fixed mini pairs (e.g. 4)
Pass	≥ 3 K points; clear diminishing Δ OOD; weak log-log r vs K
Fail	Non-nested K; steps $\not\propto$ K $\rightarrow$ small-K undertrained
Budget	~4–12 GPU-h
Script	<code>env_outline/slurm_ndemo_sweep.sh</code>

RC4 — `powerlaw_fit_smoke` (P4)

Field	Spec
Paper	Mouse pairs $y = 1.263 \cdot x^{-0.683}$ ($r=0.942$); $0.99 \Rightarrow$ 1,191 pairs
Goal	Fit $(Y=\beta X^\alpha)$ on RC1/RC2 smoke; emit prediction JSON
Pass	<code>logs/powerlaw_fit.json</code> with $\alpha < 0$, β , r; predict X for target 0.95; note 1191 unverified
Budget	CPU <1 h
Script	<code>env_outline/slurm_powerlaw_fit_smoke.sh</code> (outline in DELIVERABLE; stub if missing)

RC5 — `capacity_x_data` (P8/P9/P10)

Field	Spec
Paper	ViT 0.66/0.81/0.90; FT 0.90 / LoRA 0.72 / LfS 0.03 / frozen 0.00; U-Net 0.88/0.90/0.83
Goal	Course: ViT-S vs B, U-Net small vs base, FT vs LoRA on one mini-N card
Pass	B>S on OOD; U-Net base $\gg$ small; FT>LoRA on proxy (log MSE trap)
Defer	ViT-L, U-Net large, LfS/frozen full cells
Budget	~6–16 GPU-h; VRAM 12–24 GB
Script	<code>env_outline/slurm_capacity_x_data.sh</code> (outline in DELIVERABLE)

RC6 — (stretch) `recipe_train_smoke` (P11 partial)

Mini analogue of high-pair $\times$ $K \approx 50$ vs concentrated demos; **not** real 32×50 / Tab.1 90%. Optional P3 pairs `{1,2,4}` / P7 matched-T if GPU remains. Script outline: `env_outline/slurm_recipe_train_smoke.sh`.

Paper hyperparams to pin (App. C) + smoke cuts

Knob	Paper	Smoke cut
Batch	256	32–64
Encoder	DINOv2 ViT-L full FT	ViT-S / ResNet18; FT vs LoRA cell
U-Net	base	small vs base; defer large
Steps	up to 5e5 / 75 h 8xA800	short steps scaled to data
Nested subsets	always	keep (fairness)
Checkpoint	final	keep
Eval	8 unseen $\times$ 5 trials; human stage scores	locked OOD shard + stage-proxy / offline MSE
TE / Ah	TE=8; Ah=16 @ 5 Hz	keep if DP stack

Numbers to memorize (exam — Nykolas)

Fact	Value		
Total demos / rollouts	>40k / >15k		
Power-law $\backslash(Y\backslash)$	$\backslash(1-\text{score})\backslash)$		
PW obj/env/pair α	-0.703 / -0.844 / -0.579		
MA obj/env/pair α	-0.697 / -0.466 / -0.683		
$\backslash$	$r\backslash$	range (score laws)	0.942–0.991
MA $\rightarrow$ score 0.99	1191 pairs		
Demo-perf $\backslash(r\backslash)$ (weak)	-0.62 (PW), -0.79 (MA)		
Plateau @ 32 pairs	~1600 $\rightarrow \backslash(K=50\backslash)$		
Multi-obj @ $\backslash(M=16\backslash)$	Negligible vs 1 obj/env		
Verification SR (4 tasks)	~85–92.5%		
ViT-S/B/L	0.66 / 0.81 / 0.90		
U-Net S/B/L	0.88 / 0.90 / 0.83		
Frozen / LfS / LoRA / FT	0.00 / 0.03 / 0.72 / 0.90		
Params / max train	>396M; 5e5 steps; 75 h / 8 A800		

Failure modes cheat-sheet

Symptom	Likely cause	Next action
Flat M curve	Object-OOD used / non-nested	Eval env-OOD only; nest envs
Flat N curve	Same	Fix object holdout
No K plateau	Steps $\nless K$	Scale steps; nest K
$\alpha>0$ in fit	Used score instead of gap	Fit $\backslash(Y=1-\mathrm{score})\backslash)$
ViT-S $\approx$ B	Undertrain B	Match steps/aug
LoRA "wins"	Looking at MSE	Stage-proxy primary
Cross-run incomparable	Different eval batches	Lock one eval shard (App.E.2)
OOM	Paper bs256 / ViT-L	bs32–64; defer L
Claimed Tab.1 90% from stub	Overclaim	Report mini-N directional only
Positive α from pdftotext	Superscript minus dropped	Use Ablation Bot negative exponents

What not to invent / claim

- **Do not** invent universal multi-task / multi-emb Chinchilla laws.
- **Do not** amortize Tab.1 ~90% / Fig.5 $r>0.94$ onto Partial stubs.
- **Do not** trust MSE alone (LoRA trap).
- **Do not** claim 1191-pair verified.
- **Do not** claim course smoke = paper 8 $\times$ A800 / 15k rollouts.
- **Do not** confuse ACT temporal ensemble with ACT-as-primary-policy.
- **Do not** claim OXE-style embodiment scaling was studied.

Empiricist exam bite (Nykolas)

"Lin et al. (ICLR'25 Oral): OOD gap $\backslash(Y=1-S\approx\beta X^{\alpha})\backslash)$ with $\backslash(\alpha\sim0.5\backslash)$ to $\backslash(-0.8\backslash)$ on $\#envs/\#objects/\#pairs$ ($\backslash(r\sim0.94\backslash)$ to $\backslash(-0.99\backslash)$); demos plateau (~50/pair); recipe **32 $\times$ 50** $\rightarrow$ ~90% zero-shot; diversity $\gg$ depth; encoder scale helps, U-Net scale doesn't; Empiricist RC1–RC3 mini-N directional — never claim Chinchilla / Tab.1 parity from stubs."

6.2 Scientific Reviewer — produced despite empty seat

Claims under review

1. Single-task IL OOD generalization follows approximate **power laws** in #envs / #objects / #pairs.
2. **Diversity** $\gg$ **demo depth** once $\backslash(K\backslash)$ hits a modest plateau (~ 50).
3. Efficient recipe **32 pairs** $\times$ **50 demos** transfers across tasks of similar difficulty ($\sim 90\%$ SR).
4. Visual encoder capacity + FT matter; action U-Net width does not.

Strengths

1. Rare **real-robot** scaling study with $>15k$ rollouts and **blind** multi-policy eval.
2. Clean operationalization of OOD as **env** $\times$ **object** (not synthetic single-factor).
3. Actionable recipe validated on **two held-out tasks** with afternoon-scale collection.
4. Honest metric analysis (MSE failures; LoRA trap) and nested-subset design.
5. Fig. 5 fits with high $\backslash r \backslash$ + explicit 1191-pair prediction (even if unverified).
6. Capacity ablations (Table 2) cleanly separate vision vs action scaling.
7. Ablation Bot corpus rich despite no section titled "Ablation" (Sec.4–6+App.G).

Weaknesses / threats

1. Only **4** tasks; similar tabletop difficulty — dexterous/LH exponents unknown.
2. **6-point** fits; no $\backslash(Y_{\infty})$; extrapolation to 1191 pairs **unverified**.
3. Single algorithm (**DP**) + UMI noise floor confound "data law" vs "interface law."
4. Human scoring subjectivity mitigated by blind batches but not eliminated.
5. No task/language axis — cannot adjudicate generalist VLA scaling.
6. Selection bias: UMI-friendly tasks only (texture-poor / large occluders avoided).
7. Per-env variance (Table 12: Pour env2 **40%**) under-discussed in headline $\sim 90\%$.
8. Cross-batch incomparability (App.E.2) — replication must freeze eval shard.
9. Compute bar ($8 \times A800 \times 75$ h) limits community full-grid replication.
10. **pdftotext exponent-sign trap** — cite Ablation Bot / re-fit, not raw OCR.
11. No multi-embodiment / RL / multi-task exponent transfer shown.
12. Concurrent RUMs / generalist papers — scope carefully vs "solved robot scaling."

Verdict: Strong **empirical systems** paper with genuine scaling-law substance for the **single-task IL** regime. Do **not** over-claim as universal robot Chinchilla law. **CLEAR** for data-cluster citations on diversity budgets; **TENTATIVE** for multi-task / RL / multi-embodiment transfer of exponents.

Reviewer exam bite: Cite **Fig. 5** + **§4.3 recipe** + **Table 1**; caveat §7 (single-task, IL-only, 4 tasks, DP-only); punish "universal Chinchilla" and "course smoke = 90%" claims.

6.3 Visionary — produced despite empty seat (BEHAVIOR recipe; no Discord/Drive)

BEHAVIOR connection:

BEHAVIOR'26 stresses **many household tasks** $\times$ **house-scale scenes** $\times$ **object variation**. This paper gives the **per-primitive diversity floor**: before buying more demos of the same (scene, object), hit **~ 32 pairs** $\times$ **~ 50** . Complements **DROID scene-churn** and **UMI handheld** as the **measurement** of how much churn is enough. Aligns OXE visionary pack's **$\sim 20k$** challenge-demo framing (**TENTATIVE** global): if ~ 100 tasks, naïve equal split ~ 200 demos/task — **below** this paper's 1600 if each task is its own policy $\rightarrow$ either (a) **share** visual/action priors across tasks ($\pi_{0.5}$ / OpenPI / N1) so per-task demos can be fewer, or (b) **concentrate** 32×50 budgets on hardest contact-rich tasks and rely on transfer for easier ones.

BEHAVIOR need	Lin'25-shaped response
Scene coverage	Maximize #envs; mix cards $n_scenes / n_objects / n_pairs / K / total_N$
Object variation	Same-category OOD; 1 unique obj/env at high M
Demo budget allocation	Default ~50/pair; after plateau spend on new pairs
Long-horizon chores	Distant-history + TE + stage-wise metrics (not more demos)
Policy race	Keep diversity cards when swapping DP $\rightarrow \pi_{0.5} / OpenPI / N1$
Eval honesty	Blind multi-checkpoint scoring; freeze eval shard

CLEAR: Design prior for BEHAVIOR-like per-skill diversity budgeting.

TENTATIVE: Official BEHAVIOR'26 baseline names this recipe; exact 20k $\rightarrow$ unit mapping.

2026 BEHAVIOR data-scaling / mix recipe (9 bets)

1. Publish BEHAVIOR mix cards: $n_scenes / n_objects / n_pairs / demos_per_pair / total_N$.
2. Default ~50 **demos/pair**; spend budget on **new scenes/objects** after RC3-style plateau.
3. Gate "more data" claims with **P1/P2/P7 matched-N** Empiricist cards.
4. Scale **vision FT** (or VLM towers), not fat action heads, when data is diverse.
5. Tag `source=umi_handheld|droid_robot|oxe|behavior`; mix under shared schemas.
6. Use P4-style fits to **budget** new scenes for a target stage-SR — extrapolations = hypotheses.
7. Keep distant-history + TE + stage-wise metrics for LH chores.
8. Re-estimate laws under $\pi_{0.5} / OpenPI / N1$ — does α change?
9. Honest non-goals: course smoke $\neq$ 40k demos / 15k rollouts / Tab.1 90% / 1191 pairs.

Follow-up research

1. Repeat Fig. 5 with **ACT / π_0 -FM / VLA** heads — do exponents move?
2. Fit laws on **task count** + language (OpenVLA/Octo axis).
3. RL fine-tune scaling on top of IL diversity laws (§7 ask).
4. Multi-embodiment: does (M_{robot}) enter the same power family as (M_{env}) ?
5. Verify **1191-pair** Mouse extrapolation (or measure (Y_{∞})).
6. Dexterous / deformable / long-horizon (K) thresholds.
7. Closed-loop collector GUI: force new (env,object) before $(K>50)$.

New applications

- **Collector GUI quota:** force new (env, object) before allowing $(K>50)$.
- **Data marketplace SKU:** sell "32-pair skill packs" not "10k demo dumps."
- **Eval harness:** blind multi-checkpoint scoring as standard for scaling claims.
- **Sim BEHAVIOR:** privileged state expands pair diversity, then real $(K=50)$ polish.

Visionary one-liner: For BEHAVIOR'26, treat Lin'25 **32×50** as the **single-skill diversity unit**; spend the ~20k pool on covering units + shared generalist prior — don't deepen easy pairs.

Visionary exam bite: "For BEHAVIOR, treat Lin'25 **32×50** as the **single-skill diversity unit**; publish mix cards; scale vision FT not U-Net width; re-fit α under $\pi_{0.5}/OpenPI$ — **no Discord/Drive**."

(No Discord / Drive checklist content.)

6.4 Stakeholder — produced despite empty seat

Product one-liner: For a **single household skill**, you do **not** need 10k demos of one kitchen — you need ~32 **varied scenes/objects** $\times$ ~50 **demos** (~1.6k/skill) with a modern chunked visuomotor policy to reach ~90% zero-shot on new kitchens/objects.

Who cares

- **Data ops / robot fleet leads:** reallocates collector time from depth → scene churn (DROID-aligned).
- **BEHAVIOR / challenge teams:** per-skill floor under a ~20k total demo budget (**TENTATIVE** global from OXE pack).
- **Foundation-model labs:** explains why OXE width helps but **per-skill coverage** still needed for zero-shot without FT.
- **Hardware startups:** UMI afternoon collection → deployable Franka skill is a sales demo.
- **Non-consumers:** need certified multi-task/language gen today; no real-eval budget; require multi-emb laws this paper doesn't fit.

Buy / build decision

1. Bottleneck = **same-scene demo depth** → stop; reallocate to new (env,obj) pairs at K≈50.
2. Bottleneck = **real visual diversity / robot-in-wild** → still fund DROID/OXE; use this paper as **per-skill floor**.
3. Bottleneck = **multi-task language gen** → this paper does **not** solve it; buy OpenVLA/ π_0 -class + keep diversity cards.
4. Always budget **real stage-scored eval** (MSE lies) + env-stratified QA (Table 12 variance).

Risks: Real eval expensive (>15k rollouts); UMI SLAM / texture limits skill class; ~90% average hides **40%** environments; recipe may under-serve high-dexterity / multi-minute tasks; compute bar for full grids; overclaiming Chinchilla universality.

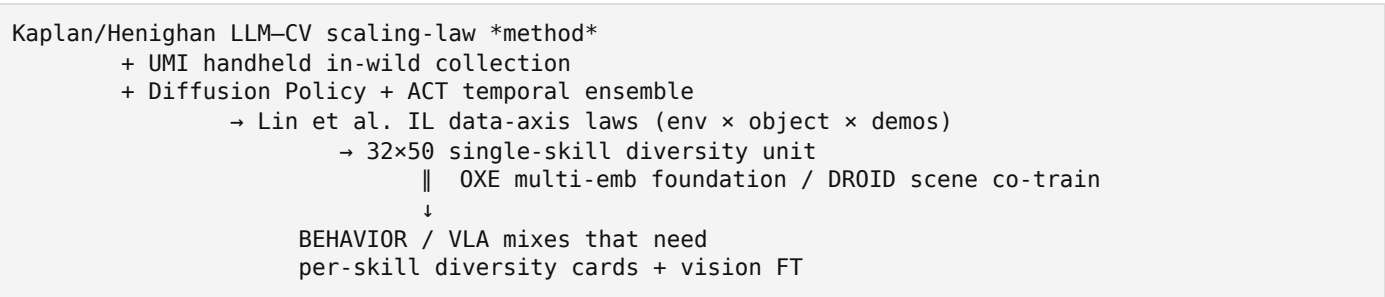
Stakeholder exam bite: "Budget **diversity pairs**, not demo depth; **32×50** is the published efficient frontier for single-task ~90% OOD — env-stratify QA; don't buy 10k same-kitchen dumps."

6.5 Archaeologist — produced despite empty seat

Priors this paper absorbs (CLEAR)

- Kaplan et al. 2020 / Henighan et al. 2020 — LLM/CV **scaling-law method** (power law on scale axis).
- **UMI** (Chi'24, 2402.10329) — collection + deploy interface.
- **Diffusion Policy** (Chi'23, 2303.04137) — policy family.
- **ACT** (Zhao'23, 2304.13705) — **temporal ensemble only** (not primary policy).
- **OXE** (2310.08864) — related data scaling; different goal (X-emb FT vs zero-shot single-task).
- **DROID** (2403.12945) — scene diversity prior; complementary.
- **DINOv2** — visual backbone.

Genealogy (CLEAR framing)



Ancestor	Relation	CLEAR/TENTATIVE
Kaplan 2020	Method analogy (power law)	CLEAR cite; exponents not transferable
Hoffmann/Chinchilla	Compute–data optimality	Not fitted here — TENTATIVE analogy only
UMI 2402.10329	Collection + deploy	CLEAR
Diffusion Policy 2303.04137	Policy family	CLEAR
ACT 2304.13705	Temporal ensemble only	CLEAR
OXE / DROID	Related data scaling; different goals	CLEAR contrast
Octo / OpenVLA / π_0	Generalist scale	Cited related; paper stops before task axis

Contrast poles (session lineage):

OXE / DROID / UMI = breadth of collection mechanisms;
Lin'25 = measurement of how much env×object diversity is enough for single-skill ZS;
Octo/OpenVLA/ π_0 = task/language scale this paper explicitly excludes.

Descendants / ecosystem

- Public code + HF processed dataset + project site — **CLEAR**.
- Conceptual influence on BEHAVIOR mix design / collector quotas — **TENTATIVE** without organizer checklist.
- Do not over-claim unique descent into VLAs without citation checks.

Genealogy one-liner: Kaplan *idea* → robot IL *env/object* laws (Lin'25 Oral); collection via UMI; DP+ACT-TE stack; **not** an OXE/Chinchilla replacement.

Archaeologist exam bite: "Lin'25 = Kaplan-style power laws on #envs/#objects/#pairs for single-task DP+UMI; K=50 plateau; 32×50 → ~90% ZS; complement OXE/DROID — do not transfer exponents to multi-task/multi-emb."

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export DSLAW_WORK_ROOT=$HOME/dslaw-smoke
cd /workspace/data-scaling-laws-quiet
bash env_outline/setup_mamba.sh
# mamba activate dslaw-scout
mkdir -p logs runs mix_cards configs
# EDIT: partition/account in slurm_*.sh
# Pin project + UMI + DP commits → logs/pins.txt
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	dslaw-scout sketch
env_outline/setup_mamba.sh	Create/update env; data/logs/mix_cards
env_outline/run_checklist.md	Ordered RC0→RC5 (+stretch)
env_outline/slurm_shard_ingest_smoke.sh	CPU ingest + mix-card gate
env_outline/slurm_env_diversity_sweep.sh	Array $M \in \{1,2,4\}$ — P1 PRIMARY
env_outline/slurm_object_diversity_sweep.sh	Array $N \in \{1,2,4\}$ — P2 PRIMARY
env_outline/slurm_ndemo_sweep.sh	Array $K \in \{10,25,50\}$ — P5 PRIMARY
env_outline/slurm_powerlaw_fit_smoke.sh	P4 fit + predict (DELIVERABLE outline)
env_outline/slurm_capacity_x_data.sh	P8/P9/P10 ViT-S/B + U-Net + FT/LoRA (outline)
env_outline/slurm_obs_horizon_lh.sh	Distant history / TE stub (outline)
env_outline/slurm_recipe_train_smoke.sh	P11 mini recipe analogue (outline)
env_outline/README.md	Cluster usage; Ablation Bot pointer; siblings

Note: Primary P1/P2/P5 scripts are present under env_outline/ ; P4/P8–P11 scripts are specified in DELIVERABLE — flesh before sbatch.

GPU / RAM / TIME budget honesty

Assumptions: scout DP (ResNet18 / ViT-S/B), image ≤ 224 , batch **32–64** (paper **256**), mini-N $M, N \in \{1,2,4\}$, $K \in \{10,25,50\}$ — **not** $M=32$ / $8 \times A800$ / $5e5$ -step.

Workload	GPU VRAM (peak)	Host RAM	Wall time (order)	Smoke
RC0 ingest	—	8–16 GB	<0.5 h CPU	Y
RC1 env M×3	10–20 GB	32–64 GB	4–14 h total	Y
RC2 obj N×3	same	32–64 GB	4–14 h total	Y
RC3 K×3	same	32–64 GB	4–12 h total	Y
RC4 power-law fit	—	8 GB	<1 h CPU	Y
RC5 capacity ×4	12–24 GB	64 GB	6–16 h total	Y
RC6/RC7 stretch	as RC1	32–64 GB	4–20 h	Partial
Paper full ViT-L bs256 @ 32×50	8×A800	large	~75 h largest	N
Paper 15k rollouts / 1191-pair	robot + mega data	—	Out of course	N

Honesty rule: log (M,N,K,encoder,ft_mode,steps,eval_shard,wall_s,metric) . If #SBATCH --gres=gpu:N but CUDA false → **fail**. Never amortize Tab.1 ~90% onto a Partial stub. Prefer **3-point mini grids** over paper {1..32} .

Course envelope: target ≤ ~40–55 GPU-h for P1+P2+P5+P4+P8/P10. Full paper grids × 4 tasks × real eval = **hundreds–thousands GPU-h + robot hours** — do not promise.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Env scale M↑ → score↑ (0.144 → 0.956 Pour)	RC1: M=4≥M=2≥M=1 env-OOD	Full M grid / paper norms
Object scale N↑ (0.133 → 0.939 Pour)	RC2: N=4≥N=2≥N=1 object-OOD	Full N grid
Demo K saturates; no clear demo power-law	RC3: diminishing returns K=10 → 50	Fig.7 plateau locations
Power-law gap vs M/N/pairs (neg. α)	RC4: α<0 fit; prediction JSON	Match Fig.5 r; 1191-pair
ViT-S/B/L 0.66/0.81/0.90	RC5: B>S on proxy	ViT-L
U-Net 0.88/0.90/0.83 null/neg	RC5: base >>> small	Include large
FT 0.90 >> LoRA 0.72	RC5: FT>LoRA on stage proxy	Frozen/LfS cells
Diversity wins @ constant total demos (P7)	Matched-T card or replot	App.G.2 full
Recipe 32×50 → ~90% SR (P11)	Mini recipe > concentrated	Real Tab.1
MSE power-law weaker (P12)	Optional; do not gate on MSE	App.G.1

Pass: Empiricist executes RC0 + **P1 RC1** + **P2 RC2** + **P5 RC3** (+ P4 fit) with **paper-aligned directional** rankings.

Fail / overclaim: "reproduced Lin'25 / matched Tab.1 90% / verified 1191 pairs / Chinchilla for robots" from mini-N stubs.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Empiricist	3 · LONGEST · ~16–18	Hypotheses, RC0–RC5, reproduce cmds, GPU honesty, ablation matrix P1–P12, exam numbers
Reviewer	empty seat · ~12–14	Fig.5 + Tab.1 + §7 caveats; verdict single-task Accept
Visionary	empty seat · ~12–14	BEHAVIOR 32×50 unit + mix cards; no Discord/Drive
Stakeholder	empty seat · ~12–14	Buy diversity pairs not depth; env-stratify QA
Archaeologist	empty seat · ~12–14	Kaplan → Lin'25 OXE/DROID; UMI+DP+ACT-TE genealogy

Spine (paste into Slide Maker) — 5 bullets:

- Diversity (envs/objects/pairs) >> demo depth** — OOD gap power-laws in $\backslash(M,N,\text{pairs})$; demos plateau.
- Must-cite:** Fig.5 exponents (PW −0.703/−0.844/−0.579; MA −0.697/−0.466/−0.683); **K=50** plateau; Table1 **85–92.5%**; ViT **0.66/0.81/0.90** vs U-Net **0.88/0.90/0.83**; frozen/LfS/LoRA **0.00/0.03/0.72**; MA → 0.99 needs **1191** pairs.
- Recipe:** $\backslash(M\{= \}N\{= \}32\backslash)$ pairs × $\backslash(K\{\approx \}50\backslash)$ (~1600 demos) → ~**90%** ZS; 1 unique obj/env; env gen harder than object.

4. **Motion/policy: DP + DINOv2 FT + ACT TE** (not classical planner / not VLA primary); Ah=16@5Hz; TE=8; vision FT >> U-Net width.
5. **BEHAVIOR'26 + honesty**: treat 32×50 as single-skill diversity unit; Empiricist RC1–RC3 mini-N; **never** invent multi-task/multi-emb Chinchilla or claim Tab.1 parity from stubs.

GitHub title sketch: [ECE 605] - Data Scaling Laws Empiricist Making Three

Full Slide Maker brief: /workspace/data-scaling-laws-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/data-scaling-laws-quiet/KANTA_PACK.md
PDF	/workspace/data-scaling-laws-quiet/KANTA_PACK.pdf
Slide brief	/workspace/data-scaling-laws-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/data-scaling-laws-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/data-scaling-laws-2410.18647-summary.md
Ablations	/workspace/papers/data-scaling-laws-2410.18647-ablations.md (AUTHORITATIVE)
Paper notes	/workspace/data-scaling-laws-quiet/paper_notes.md
Env outlines	/workspace/data-scaling-laws-quiet/env_outline/
Paper PDF	/workspace/papers/data-scaling-laws-2410.18647.pdf
Page figs	/workspace/papers/data-scaling-laws-figs/page-*.png

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 Paper Manager — Nov 23 Data · Empiricist=3 (Nycolas) LONGEST · other seats empty · all five produced · Ablation Bot AUTHORITATIVE · no Discord/Drive · no invented multi-task/multi-emb Chinchilla laws.